

Review Exam II

Complex Analysis

Underlined Definitions: May be asked for on exam

Underlined Propositions or Theorems: Proofs may be asked for on exam

Chapter 5.3

Definition. Let G be a region. A function f is meromorphic on G if . . .

Argument Principle Let G be a region and let f be meromorphic on G with poles $\{p_1, p_2, \dots, p_m\}$ and zeros $\{z_1, z_2, \dots, z_n\}$, counted according to multiplicity. Let γ be a closed rectifiable curve in G which does not pass through any of the points p_j and z_k and let $\gamma \approx 0$ in G . Then,

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz = \sum_{k=1}^n n(\gamma; z_k) - \sum_{j=1}^m n(\gamma; p_j).$$

Corollary Let G be a region and let f be meromorphic on G with poles $\{p_1, p_2, \dots, p_m\}$ and zeros $\{z_1, z_2, \dots, z_n\}$, counted according to multiplicity. Let γ be a simple closed rectifiable curve in G which is positively oriented which does not pass through any of the points p_j and z_k and let $\gamma \approx 0$ in G . Let $\text{int}(\gamma)$ denote the bounded component of $\mathbb{C} \setminus \{\gamma\}$. Let $P_{f,\gamma}$ and $Z_{f,\gamma}$ denote the cardinality of $\{p_1, p_2, \dots, p_m\} \cap \text{int}(\gamma)$ and $\{z_1, z_2, \dots, z_n\} \cap \text{int}(\gamma)$, respectively. Then,

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz = Z_{f,\gamma} - P_{f,\gamma}.$$

Corollary Let G be a region and let f be analytic on G with zeros $\{z_1, z_2, \dots, z_n\}$, counted according to multiplicity. Let γ be a simple closed rectifiable curve in G which is positively oriented which does not pass through any of the points z_k and let $\gamma \approx 0$ in G . Let $\text{int}(\gamma)$ denote the bounded component of $\mathbb{C} \setminus \{\gamma\}$. Let $Z_{f,\gamma}$ denote the cardinality of $\{z_1, z_2, \dots, z_n\} \cap \text{int}(\gamma)$. Then, $\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz = Z_{f,\gamma}$.

Problems about counting the number of zeros of a given function in a given region using Argument Principle

Rouche's Theorem (Ver #1) Let G be a region and let f and g be meromorphic on G . Let $\overline{B(a, r)} \subset G$ such that f and g have no zeros or poles on $C(a, r)$. Let Z_f, Z_g, P_f, P_g denote the cardinality of the zeros of f and g and the cardinality of the poles of f and g inside $C(a, r)$, respectively. If $|f(z) + g(z)| < |f(z)| + |g(z)|$ on $C(a, r)$, then $Z_f - P_f = Z_g - P_g$.

Corollary Let G be a region and let f and g be analytic on G . Let $\overline{B(a, r)} \subset G$ such that f and g have no zeros on $C(a, r)$. Let Z_f, Z_g denote the cardinality of the zeros of f and g inside $C(a, r)$, respectively. If $|f(z) + g(z)| < |f(z)| + |g(z)|$ on $C(a, r)$, then $Z_f = Z_g$.

Rouche's Theorem (Ver #2) Let G be a region and let f and g be meromorphic on G . Let $\overline{B(a, r)} \subset G$ such that f has no zeros or poles on $C(a, r)$. Let $Z_f, Z_{f+g}, P_f, P_{f+g}$ denote the cardinality of the zeros of f and g and the cardinality of the poles of f and $f+g$ inside $C(a, r)$, respectively. If $|g(z)| < |f(z)| < \infty$ on $C(a, r)$, then $Z_f - P_f = Z_{f+g} - P_{f+g}$.

Corollary Let G be a region and let f and g be analytic on G . Let $\overline{B(a, r)} \subset G$ such that f has no zeros on $C(a, r)$. Let Z_f, Z_g denote the cardinality of the zeros of f and g inside $C(a, r)$, respectively. If $|g(z)| < |f(z)|$ on $C(a, r)$, then $Z_f = Z_{f+g}$.

Problems about counting the number of zeros of a given function in a given region using Rouché's Theorem

Chapter 6.1

Maximum Modulus Theorem (Ver #1) Let G be a region and let $f \in A(G)$. If there exists $a \in G$ such that $|f(a)| \geq |f(z)|$ for all $z \in G$, then f is constant.

Maximum Modulus Theorem (Ver #2) Let G be a bounded region and let $f \in A(G) \cap C(\overline{G})$. Then,

$$\max_{z \in \overline{G}} |f(z)| = \max_{z \in \partial G} |f(z)|.$$

Definition. Let G be a region and let $f : G \rightarrow \mathbb{R}$. Let $a \in \partial G$ or $a = \infty$. Then, $\limsup_{z \rightarrow a} f(z) = \dots$

Definition. Let G be a region. Then, $\partial_\infty G = \dots$

Maximum Modulus Theorem (Ver #3) Let G be a region and let $f \in A(G)$. Suppose there exists a constant M such that $\limsup_{z \rightarrow a} |f(z)| \leq M$ for all $a \in \partial_\infty G$. Then, $|f(z)| \leq M$ for all $z \in G$.

Supplement Maximum Modulus

Definition (for Supplement). Let $D = \{z : |z| < 1\}$ and for $0 < r \leq 1$ let $D_r = \{z : |z| < r\}$

Theorem For $f \in A(D)$ let

$$\text{a) } M(r, f) = \max_{|z|=r} |f(z)| \quad \text{b) } A(r, f) = \max_{|z|=r} \operatorname{Re} f(z)$$

$$\text{c) } I_p(r, f) = \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(re^{i\theta})|^p d\theta$$

Then,

- (i) $M(r, f)$ is a strictly increasing function of r on $(0,1)$ unless f is constant.
- (i) $A(r, f)$ is a strictly increasing function of r on $(0,1)$ unless f is constant.
- (i) $I_p(r, f)$ is a strictly increasing function of r on $(0,1)$ unless f is constant, for $p \in \mathbb{N}$

Parseval's Identity Let $f \in A(D)$, $f(z) = \sum_{n=0}^{\infty} a_n z^n$. For $0 < r < 1$,

$$I_2(r, f) = \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(re^{i\theta})|^2 d\theta = \sum_{n=0}^{\infty} |a_n|^2 r^{2n}$$

Chapter 6.2

Let D denote the open unit disk (centered at 0) and let D_r denote open disk of radius r (centered at 0).

Schwarz's Lemma (Ver #1) Let $f \in A(D)$ such that (a) $|f(z)| \leq 1$ for all $z \in D$ and (b) $f(0) = 0$.

Then, (i) $|f(z)| \leq |z|$ for all $z \in D$ and (ii) $|f'(0)| \leq 1$. Moreover, equality occurs in (i) or (ii) if and only if $f(z) = \zeta z$ for some $\zeta, |\zeta| = 1$.

Schwarz's Lemma (Ver #2) Let $f \in A(D)$ such that (a) $f(D) \subset D$ and (b) $f(0) = 0$.

Then, (i) $f(D_r) \subset D_r$ and (ii) $|f'(0)| \leq 1$. Moreover, equality occurs in (i) or (ii) if and only if $f(z) = \zeta z$ for some $\zeta \in \partial D$.

Proposition. For $a \in D$ let $\varphi_a(z) = \frac{z-a}{1-\bar{a}z}$. Then, $\varphi_a \in A(D)$, $\varphi_a : D \rightarrow D$ and, φ_a is one-to-one and onto.

Further, for $|z| = 1, |\varphi_a(z)| = 1$ and $\varphi_a'(0) = 1 - |a|^2$, $\varphi_a'(a) = (1 - |a|^2)^{-1}$.

Proposition Let $f \in A(D)$, $f : D \rightarrow D$. For $a \in D$ let $f(a) = \alpha$. Then, $|f'(a)| \leq \frac{1 - |\alpha|^2}{1 - |a|^2}$. Equality occurs if and only if $f(z) = \varphi_{-\alpha}(\zeta \varphi_a(z))$ for some $\zeta \in \partial D$.

Theorem Let $f \in A(D)$, $f : D \rightarrow D$ such that f is one-to-one and onto. Then, there exists $a \in D$ and $\zeta \in \partial D$ such that $f = \zeta \phi_a$.

Supplement Subordination

Definition (for Supplement). Let $D = \{z : |z| < 1\}$ and for $0 < r \leq 1$ let $D_r = \{z : |z| < r\}$

Definition Let $f, g \in A(D)$. Then, f is subordinate to g ($f \prec g$) if ...

Theorem. Let $f \prec g$, $f(z) = \sum_{n=0}^{\infty} a_n z^n$, $g(z) = \sum_{n=0}^{\infty} b_n z^n$. Then,

- i) $f(0) = g(0)$, $|f'(0)| \leq |g'(0)|$
- ii) $f(D) \subset g(D)$
- iii) $f(D_r) \subset g(D_r)$
- iv) $M(r, f) \leq M(r, g)$
- v) $I_p(r, f) \leq I_p(r, g)$
- vi) $\sum_{k=0}^m |a_k|^2 \leq \sum_{k=0}^m |b_k|^2$, $m = 0, 1, 2, 3, \dots$
- vii) $\max_{z \in D_r} (1 - |z|^2) |f'(z)| \leq \max_{z \in D_r} (1 - |z|^2) |g'(z)|$

Proposition. Let $f, g \in A(D)$. Suppose (a) $f(0) = g(0)$, (b) $f(D) \subset g(D)$, (c) g is one-to-one. Then, $f \prec g$.

Example. Let $P = \{f \in A(D) : \operatorname{Re} f(z) > 0 \text{ for all } z \in D, f(0) = 1\}$. Let $p(z) = \frac{1+z}{1-z}$. Then, $f \in P$ implies $f \prec p$.

Chapter 10.1

Definition. Let G be a region and let $f : G \rightarrow \mathbb{R}$. f is harmonic on G if ...

Definition. Let G be a region and let $f : G \rightarrow \mathbb{R}$. f satisfies the Mean Value Property (MVP) on G if ...

Proposition. Let G be a region. Let f be harmonic on G . Then, f satisfies the MVP on G .

Maximum Principle (Ver. #1) Let G be a region and let $u : G \rightarrow \mathbb{R}$ satisfy the MVP on G . If there exists $a \in G$ such that $u(a) \geq u(z)$ for all $z \in G$, then u is constant.

Maximum Principle (Ver. #2) Let G be a region and let $u, v : G \rightarrow \mathbb{R}$ be bounded functions satisfying the MVP on G . Suppose for each $a \in \partial_\infty G$ that $\limsup_{z \rightarrow a} u(z) \leq \limsup_{z \rightarrow a} v(z)$, then either $u(z) < v(z)$ for all $z \in G$ or else $u \equiv v$.

Minimum Principle (Ver. #1) Let G be a region and let $u : G \rightarrow \mathbb{R}$ satisfy the MVP on G . If there exists $a \in G$ such that $u(a) \leq u(z)$ for all $z \in G$, then u is constant.

Chapter 10.2

Definition. The Poisson kernel $P_r(\theta) = \dots$

Proposition
$$P_r(\theta) = \operatorname{Re} \frac{1 + re^{i\theta}}{1 - re^{i\theta}} = \frac{1 - r^2}{1 - 2r \cos(\theta) + r^2} = \operatorname{Re} \frac{e^{i\theta} + r}{e^{i\theta} - r}$$

Proposition. The Poisson kernel satisfies

- (i) $\frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(\theta) d\theta = 1, \quad 0 < r < 1$
- (ii) $P_r(\theta) > 0, \quad P_r(\theta) = P_r(-\theta)$
- (iii) $P_r(\theta) < P_r(\delta)$ for $0 < \delta < |\theta| < \pi$, i.e., $P_r(\theta)$ is strictly decreasing on $(0, \pi)$
- (iv) $\lim_{r \rightarrow 1^-} P_r(\delta) = 0$ for each $\delta, 0 < \delta \leq \pi$

Theorem Let $f \in C(\partial D)$, $f : \partial D \rightarrow \mathbb{R}$. Then, there exists $u \in C(\overline{D}) \cap \operatorname{Har}(D)$ such that

- (i) $u(e^{i\theta}) = f(e^{i\theta})$
- (ii) u is unique

Corollary Let $u \in C(\overline{D}) \cap \operatorname{Har}(D)$. Then, $u(re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(\theta - t) u(e^{it}) dt, \quad 0 < r < 1$

Corollary Let $h \in C(C(a, R))$, $h : C(a, R) \rightarrow \mathbb{R}$. Then, there exists $w \in C(\overline{B(a, R)}) \cap \operatorname{Har}(B(a, R))$ such that $w(z) = h(z)$ on $C(a, R)$.

Corollary Let $u \in C(\overline{B(a, R)}) \cap \operatorname{Har}(B(a, R))$. Then, $u(a + re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_{r/R}(\theta - t) u(a + re^{it}) dt, \quad 0 < r < R$

Harnack's Inequality Let $u \in C(\overline{B(a, R)}) \cap \text{Har}(B(a, R))$, $u(z) \geq 0$. Then,

$$\frac{R-r}{R+r}u(a) \leq u(a+re^{i\theta}) \leq \frac{R+r}{R-r}u(a), \quad 0 \leq r < R$$

Chapter 7.1

Definition. Let G be a region and let (Ω, d) be a complete metric space. Then, $C(G, \Omega) = \dots$

Proposition. Let G be a region. Then there exists a sequence of subsets $\{K_n\}$ of G such that

- (i) $K_n \subset\subset G$
- (ii) $K_n \subset \text{int}(K_{n+1})$
- (iii) $\bigcup_{n=1}^{\infty} K_n = G$
- (iv) $K \subset\subset G$ implies $K \subset K_n$ for some $n \in \mathbb{N}$

Lemma If (S, d) is a metric space, then (S, σ) is a metric space, where $\sigma(s, t) = \frac{d(s, t)}{1 + d(s, t)}$. A set O is open in (S, d) if and only if O is open in (S, σ) .

Definition. For $K \subset\subset G$ and $f, g \in C(G, \Omega)$, let $\rho_K(f, g) = \dots$, $\sigma_K(f, g) = \dots$, $B_{\rho_K}(f, \delta) = \dots$.

Definition. For $\{K_n\}$ a compact exhaustion of a region G and for $f, g \in C(G, \Omega)$ let $\rho(f, g) = \dots$

Proposition. $(C(G, \Omega), \rho)$ is a metric space.

Lemma 1.7 (i) Given $\varepsilon > 0$ there exists $\delta > 0$ and $K \subset\subset G$ such that for $f, g \in C(G, \Omega)$

$$\rho_K(f, g) < \delta \Rightarrow \rho(f, g) < \varepsilon$$

(ii) Given $\delta > 0$ and $K \subset\subset G$ there exists $\varepsilon > 0$ such that

$$\rho(f, g) < \varepsilon \Rightarrow \rho_K(f, g) < \delta$$

Lemma 1.10 (i) A set $O \subset C(G, \Omega)$ is open if and only if for each $f \in O$ there exists $\delta > 0$ and $K \subset\subset G$ such that $O \supset B_{\rho_K}(f, \delta)$

(ii) A sequence $\{f_n\} \subset C(G, \Omega)$ converges to f (in the ρ metric) if and only if for each $K \subset\subset G$ $\{f_n\}$ converges to f in the ρ_K metric.

Proposition $(C(G, \Omega), \rho)$ is a complete metric space.

Definition. A set $F \subset C(G, \Omega)$ is normal $\dots$

Proposition. A set $F \subset C(G, \Omega)$ is normal if and only if $\overline{F}$ is compact.

Proposition. A set $F \subset C(G, \Omega)$ is normal if and only if for each $\delta > 0$ and $K \subset\subset G$ there exist functions

$$f_1, f_2, \dots, f_n \in F \text{ such that } F \subset \bigcup_{k=1}^n B_{\rho_K}(f_k, \delta).$$

Definition. A set $F \subset C(G, \Omega)$ is equicontinuous at a point $z_0 \in G$ if ...

Definition. A set $F \subset C(G, \Omega)$ is equicontinuous on a set $E \subset G$ if ...

Proposition. Suppose a set $F \subset C(G, \Omega)$ is equicontinuous at each point of G . Then, F is equicontinuous on each $K \subset\subset G$.

Arzela-Ascoli Theorem A set $F \subset C(G, \Omega)$ is normal if and only if

- (a) for each $z \in G$ the orbit of z under F , i.e., $\{f(z) : f \in F\}$, has compact closure and
- (b) $F \subset C(G, \Omega)$ is equicontinuous at each point of G .